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# CSU Senior Project: Discreet Dispatch

By: Andrew Burbage

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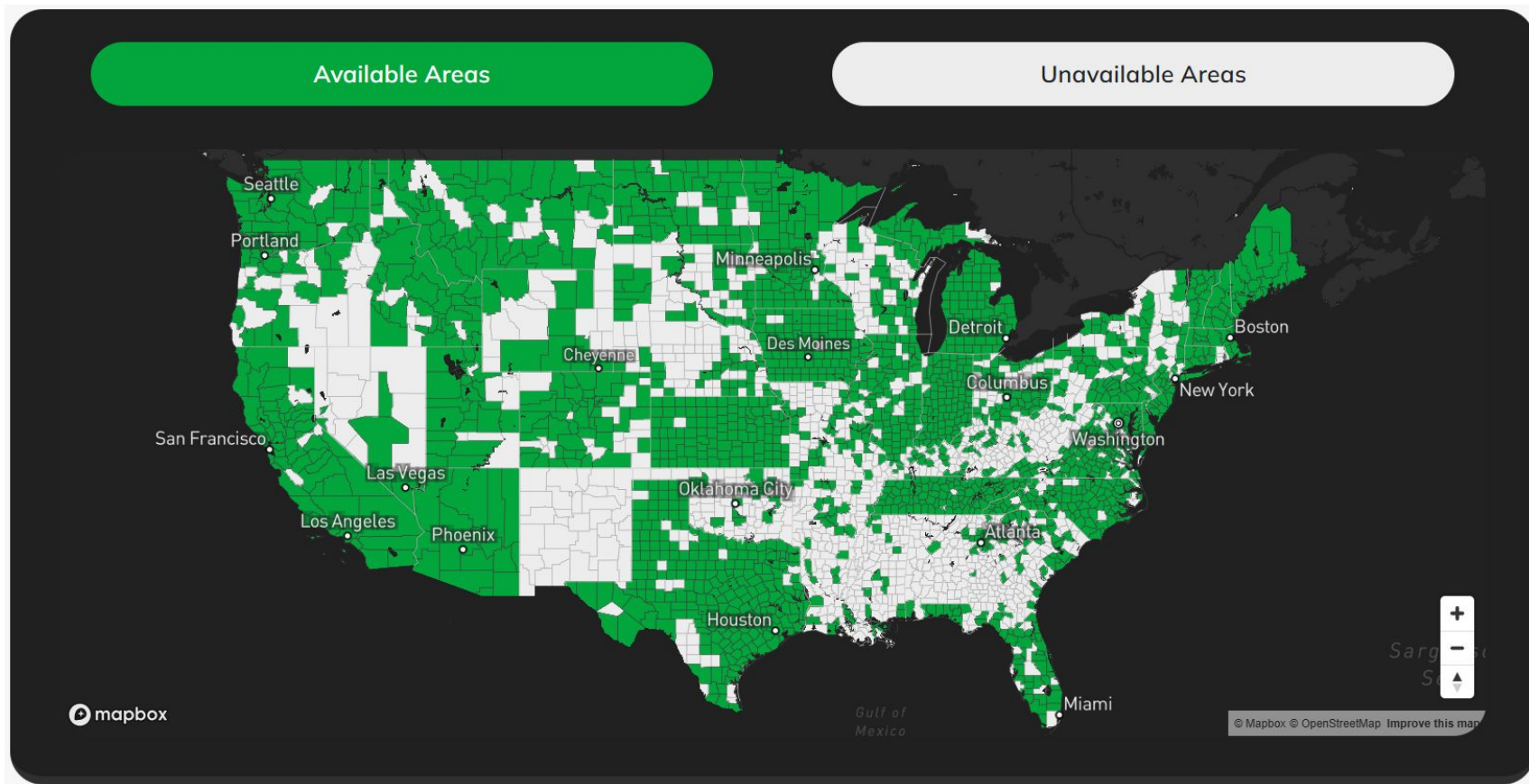


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## Problem

The ability to call 911 is taken for granted.

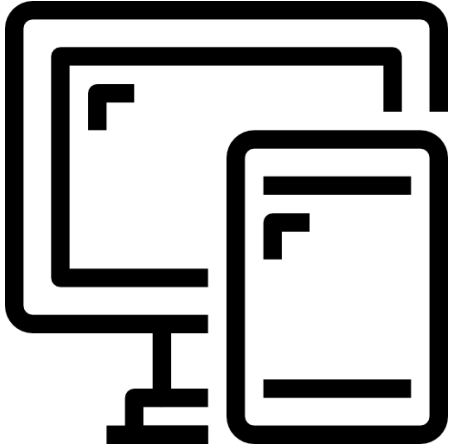
- FCC requires carries to offer E911 Services but not all dispatch can handle them
- Currently only 61% of dispatch centers provide such services (source: [text911.info](http://text911.info))
- Those who are not part of that fortunate 61% are left in the dark



Map of Current E911 coverage (Sourced from [text911.info](https://text911.info) on 4/14/26)

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# Possible Solutions

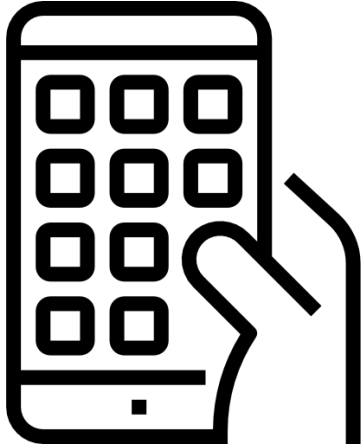


## Mobile Based

- Easier to access in emergencies
- May be hard to reach if phone is not available
- More challenging to develop

## Web Based

- More Accessible device wise
- May be hard to use under duress
- Easier to develop



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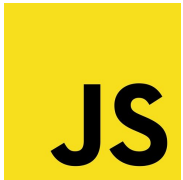
# Solution

## Mobile Based

- Easy to get to and use in an emergency
- Shows more skill and allows for learning important field

## How it works

- Allows non-E911 dispatch centers to get a phone call
- User fills out survey that forms script that is read by telephony service



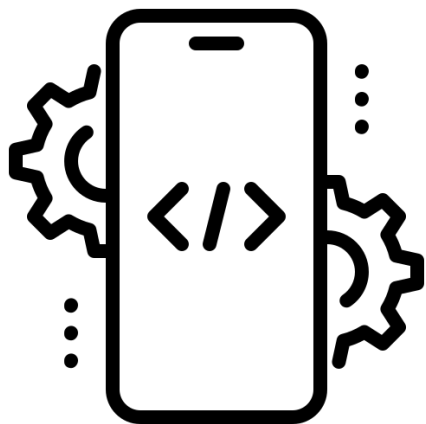
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# Tech Stack

**Languages:** Kotlin and Javascript

**Software:** Android Studios, Compose, Google Materials, Cloudflare Workers, and Twilio Telephony Services

**Hardware:** Laptop for building and Android phone for use

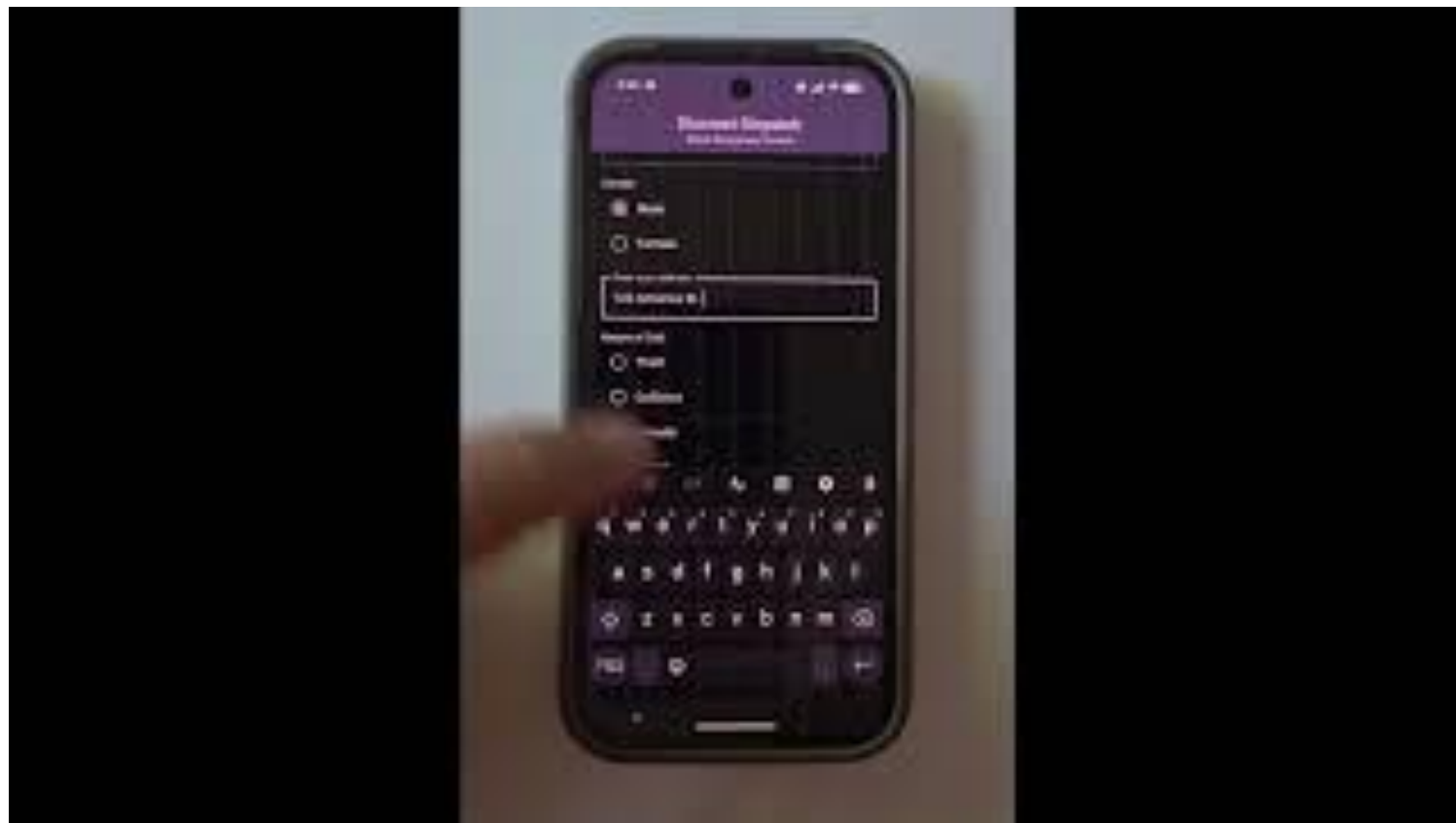


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## Process

- Design
  - Flow
  - Means
- Implimentation
  - Layouts
  - Setups
- Testing
  - Individual parts to the whole application

# Demo



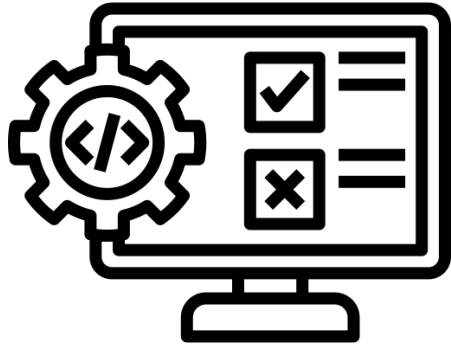


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# Requirements



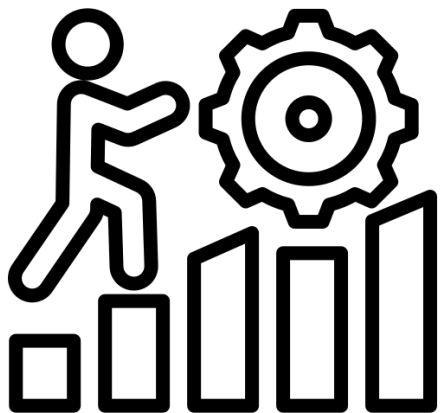
- Functionality
- Look & Feel
- Usability
- Performance
- Maintainability & Support
- Security



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# Testing

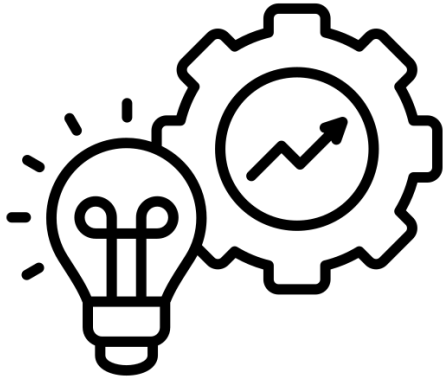
- Unit
  - Data (Collection and Integrity)
  - Functions functioning
- Integration
  - App flow
  - Call flow
- Aesthetics
  - UI/UX
  - Study Group



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## Challenges

- FCC and TTS
  - The Government has a lot of opinions (regulations)
- XML vs Compose
  - Function? UI? ... or Both?
- One Drive File Lock
  - Android Studio likes its files



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## Future Enhancements

- GPS based Location Services (Google Maps or ATAK-Civ)
  - Location is critical to emergency response
- Designated Database for Call Logs
  - Troubleshooting and Subpoenas
- Location Based Calling (Not just the test phone number)
  - Routing calls

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# Sources

accessSOS. (2022). Text to 911. <https://text911.info/>

FCC. (2020, January 6). *Text to 911: What you need to know*. Federal Communications Commission.  
<https://www.fcc.gov/consumers/guides/what-you-need-know-about-text-911>

Yan, H. (2019, November 26). *A 911 call with a fake pizza order helped stop a possible attack. but what if you can't speak to 911?* CNN.  
<https://www.cnn.com/2019/11/26/us/what-to-do-if-you-cant-speak-to-911/index.html>

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